MIHIR SHARMA

Designer turned HCI researcher · Haptics · Accessibility · Human–AI interaction
msharma.des@gmail.com · https://msharma.framer.website/

SUMMARY

I trained as a product designer and worked in visual and UX design, including a General Motors–sponsored autonomous-vehicle project. That work led me to one research question: how should a machine that perceives the world for a blind person speak to them through touch, and how should it admit when it is unsure? I began with a built system tested in a climbing gym (TactileGrip, my MFA thesis). In 2026 I turned the same question toward AI. Three papers are now under review at ACM IUI, CHIIR, and PACM IMWUT.

Research interests: haptic and multisensory interfaces · accessibility for blind and low-vision people · calibrated trust in AI · embodied human–AI interaction

EDUCATION

MFA, User Experience Design, with Distinction, College for Creative Studies, Detroit
2022 – 2024
GPA 3.885/4.0 · President's List every term · Gold Medalist
Thesis: TactileGrip: A Multi-sensory Climbing Interface for Visually Impaired Athletes (Thesis Select Award)
Coursework: Graduate Research Methods, Human Factors, Artificial Intelligence, UX Research, Rapid Prototyping

Postgraduate Program, User Experience Design, Indian Institute of Technology Bombay
2022

B.Des, Product Design, University of Petroleum and Energy Studies, India
2016 – 2020

PUBLICATIONS AND SUBMISSIONS

Under review

- Mihir Sharma. Can a Buzz Answer That? How Much of What Blind People Ask a Visual Assistant Fits a Vibration Vocabulary. ACM IUI 2027 Posters. Of 8,000 held-out VizWiz questions, 15.5% could be encoded as a vibration rhythm and 26.8% more as a tactile "can't tell". A real VQA model (ViLT) gave a confident answer on 20% of the photos the crowd marked unanswerable. Ships with Felt Answers, an offline phone prototype with identification-trial and device-check modes.
- Mihir Sharma. Material Confidence: Which Outcomes Can Tell Felt from Seen AI Confidence? ACM CHIIR 2027 Short Papers. Re-analysis of 35,670 human–AI decisions from 1,125 people, plus a simulation-based design analysis. With 96 participants, reliance sensitivity detects a haptic confidence cue 99% of the time. The common appropriate-reliance measure would need about 1,820 participants.
- Mihir Sharma. TrueTap: Budgeting How Often a Smartwatch Buzzes You by Mistake. PACM IMWUT. A calibration method with a statistical guarantee. On day-long wrist data from 151 adults, it cut the share of wearers over a false-buzz budget from 7.6% to 3.7%, within a 9.8% bound.

In preparation (WIP)

- Mihir Sharma. Feel the Frame: Touch-Guided Camera Aiming for Blind Users of Visual AI Assistants. An error-recovery policy that gives one correction at a time and gives up gracefully. Five image-flaw detectors, trained and tested on 7,750 VizWiz photos, reach held-out AUROC .66–.95. Includes a designed four-motor sleeve with host-tested firmware.
- Mihir Sharma. SenseShift: Adaptive AI-to-Haptic Communication for Blind and Low-Vision People. Three studies with BLV adults fully specified, with study apps and an analysis pipeline. In simulation, the attention policy cut distractor messages from 58.9% (nearest-object baseline) to 6.2%.

RESEARCH EXPERIENCE

Independent Researcher, Haptic and embodied human–AI interaction · 2026–present

- Wrote three sole-authored papers for different ACM venues in one year. Each comes with a reproducible research package, automated number checks and clean-rebuild verification.
- Designed and built research prototypes:
  - *Felt Answers*, an offline phone app with identification-trial and device-check modes
  - *Feel the Frame*, an Android camera-aiming app plus a four-motor sleeve design with host-tested firmware
  - *SenseShift*, study apps, band firmware and an analysis pipeline

Material Sustainability Assistant, Ecosphere Organics, Detroit · 2025–2026

- Researched turning agricultural bio-waste (banana peels, corn husks, orange pulp, coffee grounds) into biodegradable composites for design and HCI.
- Explored composites with embedded micro-sensors and reactive pigments that sense moisture, decay, and temperature.

TactileGrip: climbing holds that guide blind climbers, MFA thesis, College for Creative Studies, with Predrag K. Nikolić · 2023–2024

- **Problem:** blind climbers depend on a guide calling out each hold from the ground, a channel that fades with height and gym noise.
- **Research:** ran 20 interviews with BLV climbers, guides, coaches and gym staff, plus an autoethnographic climb. In a workshop, all 10 blindfolded expert climbers froze on routes they knew by memory.
- **System:** Arduino holds with an ultrasonic beep that quickens as the hand approaches, a vibration that confirms the grip, 3D-printed forms that point to the next hold, and checkpoint holds that announce progress aloud.
- **Testing:** ergonomics testing cut two of three hold forms; two rounds of voice-only usability testing on the companion app.
- **Result:** every climber completed more of the route than with verbal guidance alone ($p = .002$), and guides gave no directional instructions during hold-level navigation.

DetroTune: sound, space and cultural memory in VR · 2023

- Built a Unity VR environment that teaches cultural history through spatial sound and motion.
- In a mixed-methods study ($N = 40$), participants showed higher memory retention and engagement than with video-based learning.
- Exhibited at Ars Electronica Festival and Detroit Month of Design, 2023.

NewTune: adaptive keyboard for learners with ADHD · 2022

- Built a keyboard that turns each keystroke into a musical tone and colour feedback (Python, C#).
- Preliminary trials ($N = 15$) suggested higher attention and fewer typing errors.

DESIGN AND INDUSTRY EXPERIENCE

UX Designer, Vangard, General Motors–sponsored future-mobility studio, College for Creative Studies, Detroit · 2024

- Led UX for a Level 4 autonomous-vehicle concept for Chongqing in 2040, working with exterior and interior designers. Owned system behaviour, transitions, edge cases, and warnings.
- Concept shortlisted by GM. One of two teams invited to refine it with GM designers over the summer, producing 45+ infotainment wireframes.
- Designed proactive route-intent messaging, a persistent automation-state indicator, and override controls. This work became the paper *Intent Through Touch*.

UX prototype collaboration, Ford Motor Company × HFLI, Detroit · 2023

- Led co-design workshops to set usability priorities and mentored junior contributors.
- Applied accessibility, ergonomics, and ADA principles to mobility prototypes.

Farmacy, design project · 2023

- Designed a farming app that brings field mapping, crop recommendations, and profit forecasting into one place.
- Nominated for the UX Design Awards, Berlin, 2023.

UX Researcher, Oeson Global, India · 2022

- Led two designers in redesigning a training website with high bounce rates, based on interviews and usability tests with enrolled users and drop-offs. Shipped with the development team.

Visual Designer, Helping Hand International, India · 2020–2021

- Built and led a core UI/UX team; the website redesign cut bounce rate by 17%.
- Designed assets for the Youth Empowerment Summit 2021; registration rose 73%.

TEACHING

College for Creative Studies – Teaching Assistant – Detroit
2022 – 2025

- Taught colour theory, typography, layout, and visual hierarchy to high school and undergraduate students; guided portfolio development.
- Gave one-on-one support to students with special needs.

Grosse Pointe Artists Association – Visiting Instructor – Detroit
2024

- Led workshops on designing for the senses, translating tactile and auditory cues into design tools.

SKILLS

- *Research methods:* experimental design, simulation-based power analysis, semi-structured interviews, co-design workshops, autoethnography, usability testing, mixed methods, accessibility audits (WCAG 2.2)
- *Analysis and programming:* Python, JavaScript, C#; signal-detection analysis, logistic regression
- *Physical prototyping:* Arduino, ESP32; ultrasonic, vibrotactile and optical pulse sensors; 3D printing; Blender
- *XR and interaction design:* Unity, Unreal Engine, spatial audio, Figma, Framer, ProtoPie, Adobe CC

SERVICE, TALKS AND EXHIBITIONS

- Reviewer, ACM AutomotiveUI 2026 (two papers)
- Reviewer, Ubicomp/ISWC 2026 UbiComp 2026 Poster (three papers)
- Invited talk, Scarab Club Symposium on AI and the Arts, Detroit, 2024
- Exhibitor, Ars Electronica Festival, Linz, Austria, 2023 (DetroTune)
- Exhibitor, Detroit Month of Design, 2023 (DetroTune)

AWARDS AND HONORS

- Thesis Select Award, College for Creative Studies, 2024 (TactileGrip)
- UX Design Award nominee, Berlin, 2024 (TactileGrip)
- Gold Medalist, College for Creative Studies, 2024
- UX Design Award nominee, Berlin, 2023 (Farmacy)
- President's List, College for Creative Studies, every term, 2022–2024